

Game Size vs. Success: Analyzing Video Game Install Sizes, Reviews, and Player Engagement

Ayden Trusty

Bellevue University

DSC680

Professor Amirfarrokh Iranitalab

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Business Problem

Modern video games continue to increase in size as developers push graphical fidelity, open-world environments, high-resolution textures, downloadable content, and online functionality. Many modern AAA games now require over 100 GB of storage space, which raises questions about whether larger games actually lead to stronger player engagement, better reviews, or increased popularity. This project explores whether there is a relationship between video game install size and measures of player reception such as review ratios, recommendations, Metacritic scores, and peak concurrent users.

Background and History

Video game install sizes have increased significantly over the past two decades. Earlier PC and console games often required only a few gigabytes of storage, while many modern titles now exceed 100 GB. Improvements in graphics, voice acting, open-world environments, and downloadable content have all contributed to these increases. Digital distribution platforms such as Steam now provide large amounts of public metadata related to reviews, recommendations, and player engagement, making it possible to analyze how install size may relate to popularity and reception.

Data Explanation

This project combines publicly available Steam datasets with install size information gathered through the Steam Store API. The primary dataset included review counts, peak concurrent users, recommendations, genres, playtime statistics, and Metacritic scores. Additional install size information was extracted from Steam system requirement fields using Python regular expressions. Several preprocessing steps were performed including duplicate

removal, date conversion, handling missing values, and extracting install size values from semi-structured HTML text returned by the Steam API.

Methods

Python and Jupyter Notebook were used throughout the project. Pandas was used for cleaning and transformation, while Matplotlib and Seaborn were used for visualization. Exploratory data analysis methods including scatterplots and correlation matrices were used to identify relationships between install size and measures of player engagement and review performance.

Analysis

The analysis showed that install size alone is not a strong predictor of overall game quality or player satisfaction. While some large AAA games demonstrated high player engagement and recommendation counts, smaller games were also capable of receiving extremely strong review scores and positive player reception. The results suggest that gameplay quality, optimization, replayability, and community support likely contribute more heavily to long-term success than overall install size alone. However, install size did show a moderately positive relationship with peak concurrent users, suggesting that larger games may still attract larger active player populations.

Assumptions

This project assumes that publicly available Steam metadata reasonably reflects player engagement and game popularity. It also assumes that extracted storage requirement values

closely approximate actual install sizes, although updates and downloadable content may alter storage requirements over time.

Limitations

Several limitations affected the project. Steam metadata was occasionally incomplete or inconsistent, especially among smaller games. Install size extraction required parsing semi-structured text and HTML formatting, which introduced the possibility of missing values and extraction errors. Additionally, the analysis focused primarily on PC gaming through Steam and may not fully represent other gaming platforms.

Challenges

One of the biggest challenges involved extracting install size information from Steam metadata. Storage requirements were embedded inside HTML-formatted system requirement text rather than provided as clean numerical fields. Additional debugging and preprocessing were required to correctly isolate storage requirement values.

Future Uses and Additional Applications

This project could be expanded by incorporating console gaming datasets, sales databases, or additional APIs. Future work could also analyze relationships between install size and genres, downloadable content, development budgets, or long-term player retention.

Recommendations

The results suggest that developers should avoid assuming that larger install sizes automatically improve player satisfaction or popularity. Smaller and more optimized games were still capable of achieving strong player reception and engagement. Developers may benefit more from focusing on gameplay quality, optimization, and replayability rather than simply increasing game scale.

Implementation Plan

The workflow involved loading and cleaning gaming datasets, enriching the data using the Steam Store API, extracting install size values, and merging the results into a unified analysis dataset. Exploratory visualizations and statistical comparisons were then used to analyze relationships between install size and engagement metrics.

Ethical Assessment

This project relied entirely on publicly available metadata and did not involve private user information. However, public review systems may still contain biases related to review bombing, platform-specific communities, or temporary popularity spikes.

Conclusion

This project explored whether larger video game install sizes correlate with stronger player engagement, reviews, and popularity. The analysis demonstrated that while larger games sometimes attract larger player populations, install size alone does not guarantee stronger reviews or critical reception. Smaller games were still capable of achieving high review scores

and long-term popularity, suggesting that gameplay quality and optimization remain more important than overall game scale alone.

Visual Analysis

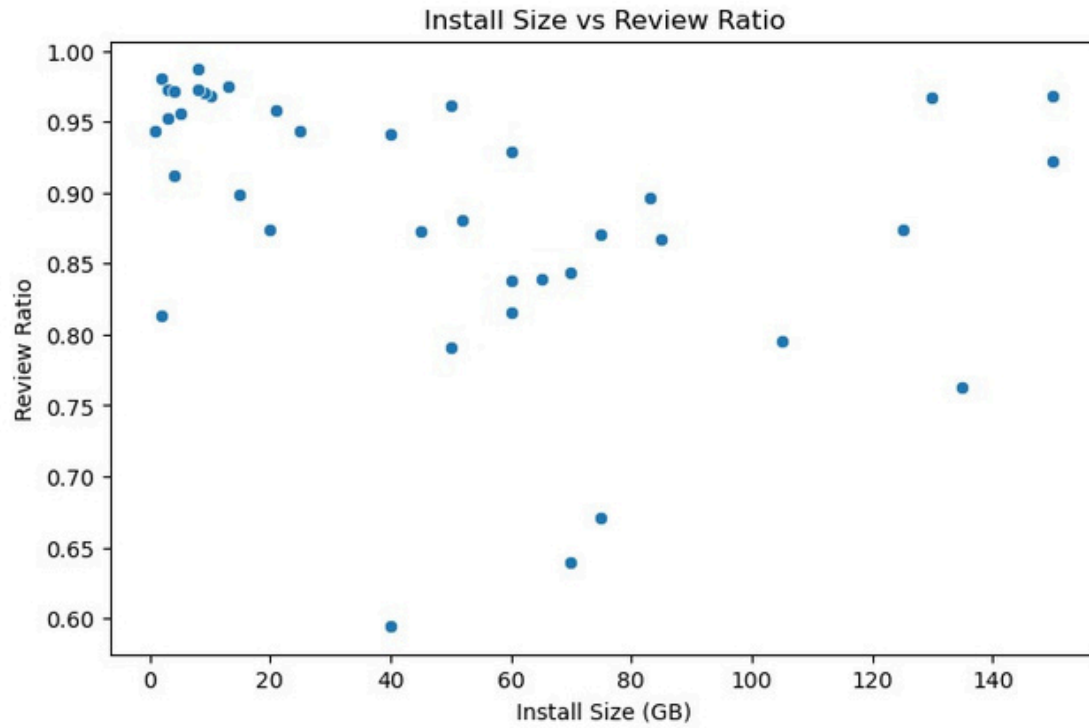


Figure 1. Install Size vs Review Ratio.

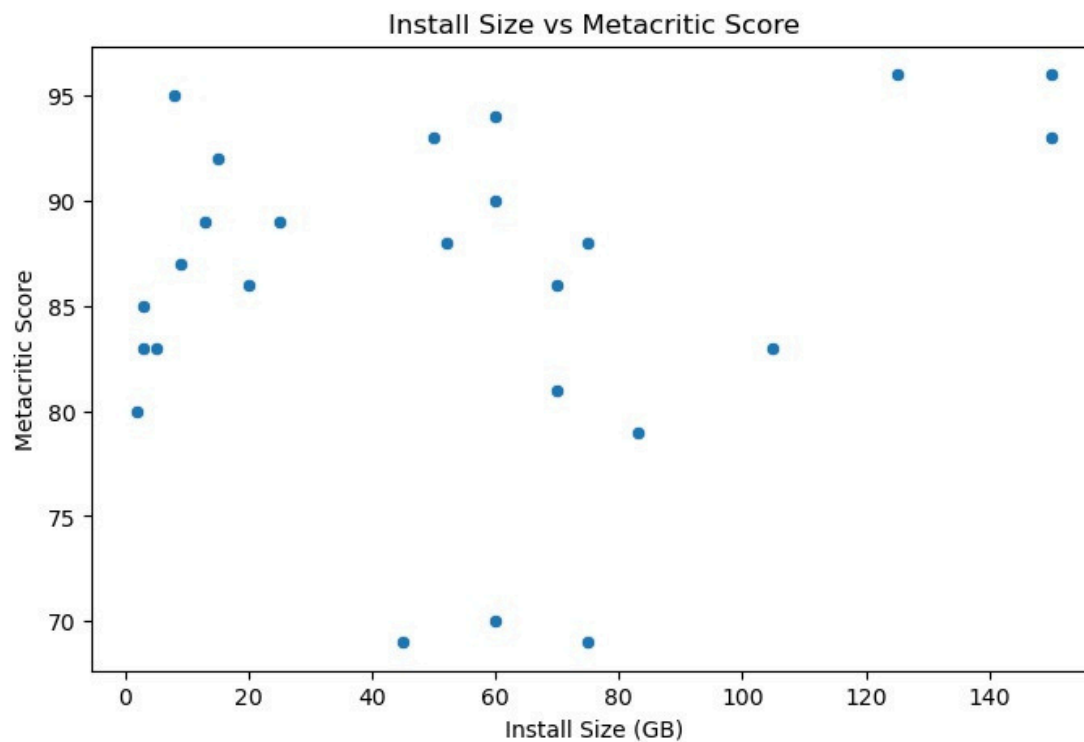


Figure 2. Install Size vs Metacritic Score.

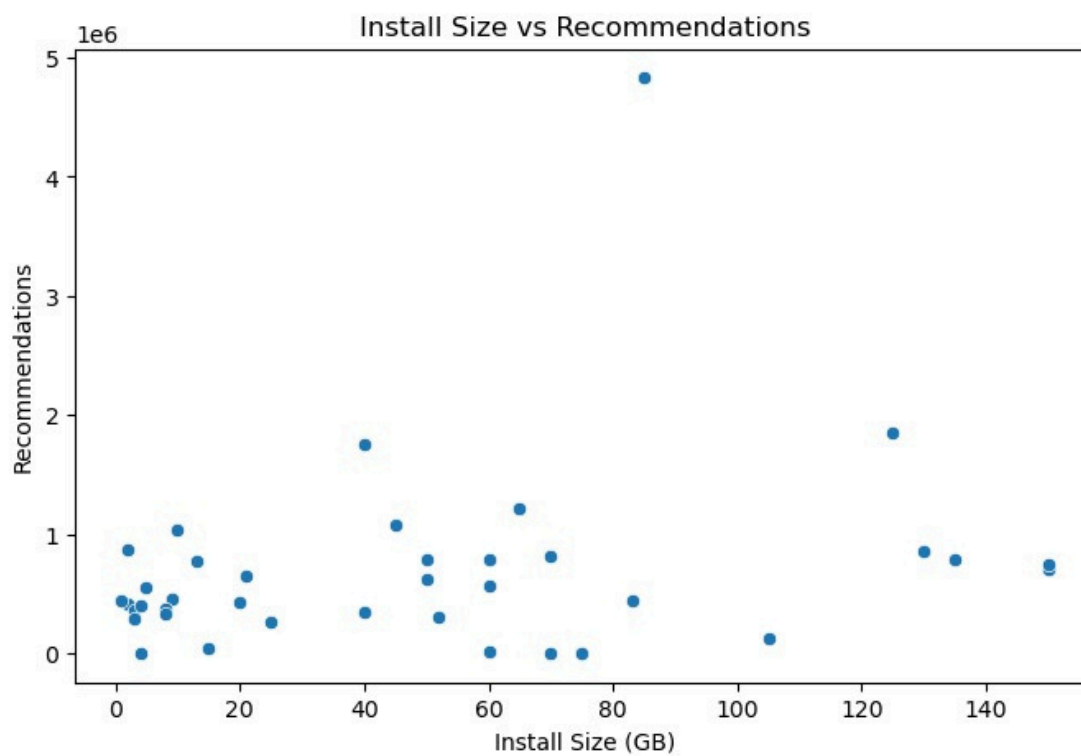


Figure 3. Install Size vs Recommendations.

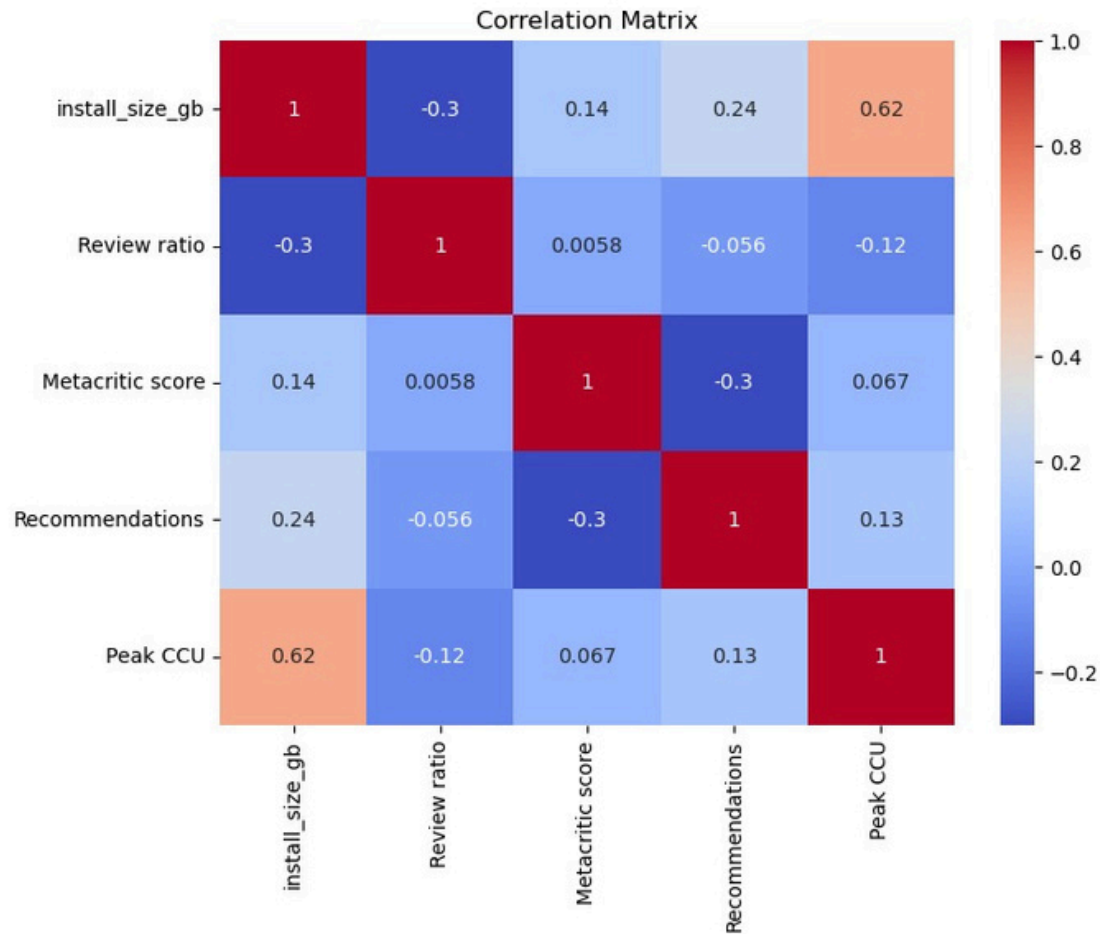


Figure 4. Correlation Matrix.

References

Kaggle. (n.d.). Steam games dataset. <https://www.kaggle.com/datasets/fronkongames/steam-games-dataset>

Kaggle. (n.d.). Video game sales with ratings.

<https://www.kaggle.com/datasets/rush4ratio/video-game-sales-with-ratings>

Steam. (n.d.). Steam store API. <https://store.steampowered.com/api/>

Valve Developer Community. (n.d.). Steam web API.

https://developer.valvesoftware.com/wiki/Steam_Web_API

RAWG. (n.d.). RAWG video game database API. <https://rawg.io/apidocs>

Appendix

The appendix contains the primary Jupyter Notebook workflow used during the project including API extraction, preprocessing, and visualization generation.

10 Questions an Audience Might Ask

- Why was Steam chosen as the primary platform?
- How accurate are the install size values?
- Why were some games missing install size data?
- Does install size vary by genre?
- Could updates change install size values?
- Why use engagement metrics instead of sales figures?
- Would console data change the results?
- How could review bombing affect the analysis?
- Could machine learning improve future versions of the project?
- What additional variables could strengthen the analysis?

DSC680 Ayden Trusty Week 10 Milestone 2 Notebook

May 24, 2026

0.1 GameSizevs.Success

0.2 AnalyzingVideoGameFileSizes,Sales,andReviewsOverTime

Project Goal: Explore whether larger video games tend to sell more and receive better review scores. Are they worth the space they take up on your hard drive?

```
[20]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import requests
import time
import re

# Optional later
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score

pd.set_option("display.max_columns", None)
```

```
[9]: # Kaggle datasets
steam_df = pd.read_csv("games.csv")
sales_df = pd.read_csv("Video_Games_Sales_as_at_22_Dec_2016.csv")

steam_df.head()
```

```
[9]:
```

	AppID	Name	Releasedate
25394	BlackDragonMagePlaytest	Aug1,2023	0-0
30	SupiparaChapter1Springas Com	Jul29,2016	0-20000
49635	MysterySolitaireTheBlack!	May6,2019	0-20000
0	██████VtubeParanoid	Oct31,2024	0-20000
10344	MazeQuestVR	Apr24,2025	0-20000
00			
32921	Estimated ownPeak CCR	Required age	PriceDiscountDLC count
20394	0	0	0.0 0 0
30310	0	0	0 65 0
49635	0	0	5.2 0 0
0	1	0	4 0 1
10344			4.9
00			9
32921		1	8.9
90			9

3631080	0	0	4.99	0	0
---------	---	---	------	---	---

About the game

2539430

NaN

496350SpringtimeApril:whenthecherrytreescome...

1034400 Immerse yourself in the most beloved, mystical...

3292190 synopsis 'Hello, I'm Hiyoro, a new YouTuber!' ...

3631080 Its not just a Maze; its a Quest! Enter the ca...

Supported languages Full audio languages

25394

[]

[]

30

['English']

[]

49635 ['English','French','German','Russian']

[]

0

['Korean']

['Korean']

10344

['English']

['English']

00

32921 Reviews

Headerimage

25394 N https://shared.akamai.steamstatic.co

30310 a m/store_it...

49635 N https://shared.akamai.steamstatic.co

0 N m/store_it...

10344 a https://shared.akamai.steamstatic.co

00 N m/store_it...

32921 N https://shared.akamai.steamstatic.co

2539430 496350/store_it... NaN

3031000 N http://mangagamer.org/steamstatic.co

3292190 N http://www.facebook.com/8FloorGames/

https://www.realityexpanded.com/bookNaN

games N

N

a

Supporturl

Supportemail \

25394 N

NaN

NaN

30 http://mangagamersupport@mangagamer.com

49635 https://www.facebook.com/8FloorGames/ support@8floor.net

0 NaN yujingamesc@gmail.com

10344 https://www.realityexpanded.com/support@realityexpanded.com

00

32921 Windows Mac Linu Metacritic score Metacritic ur Userscore \

2539 Tr False x 0 NaN 0

30310 ue False Fals 0 NaN 0

4963 Tr True e 0 NaN 0

50 ue False Fals 0 NaN 0

1034 Tr False e 0 NaN 0

400 ue Fals

3292 Positive Negative Scorerank Achievements Recommendations

2539430 ue 0 Fals NaN 0 0

496350 T252 e3 NaN 0 231

080 ue Fals

e

1034	2	3	Na	0	0
400	1	0	N	19	0
3292	0	0	Na	0	0
190	0		N		
3631			Na	Notes\	
0809			N	N	
430				a	
4963				N	
50	The game includes the following elements. 1. G...			N	
1034				NaN	
400				N	
3292	Average playtime for		Average playtime	two weeks	
2909		0		a	0
4001		8		N	0
0903		0			0
50		0			0
1034		0			0
400					
3292	Median playtime for		Median playtime	two weeks	
2909		0			0
4001		8			0
0903		0			0
50		0			0
1034		0			0
400					
3292	Developers		Publishers		
2909	NaN		NaN		
4001	minori		MangaGamer		
0903	SomerGames		8floor		
50	□□□□□		□□□□□		
1034	Reality Expanded Re		Reality Expanded LLC		
400					
3292				Categories\	
2909				NaN	
4001	Single-player,Steam Trading Cards,Steam Cloud,...				
0903	Single-player,Family Sharing				
50	Single-player,Steam Achievements,Family Sharing				
1034	Single-player,VR Only,Steam Leaderboards,Famil...				
400					
3292	Genres\				
2909	NaN				
4001	Adventure				
0903	Casual				
50	Casual,Indie,Simulation				
1034	Action,Early Access				
400					
3292				Tags \	
190					
3631					
080					

```

2539430 NaN
496350 Adventure, Visual Novel, Anime, Cute
1034400 Casual, Card Game, Solitaire, Puzzle, Hidden Object...
32921 NaN
90 NaN
36310
80 Screenshots Movies
2539430 https://shared.akamai.steamstatic.com/store_it...
496350 https://shared.akamai.steamstatic.com/store_it.a
1034400 https://shared.akamai.steamstatic.com/store_it...
3292190 https://shared.akamai.steamstatic.com/store_it...
3631080 https://shared.akamai.steamstatic.com/store_it...

```

```
[5]: sales_df.head()
```

```

[5]:
      Name Platform Year_of_Release Genre Publisher
0    WiiSports  Wii         2006    Sports Nintendo
1  SuperMario Bros. NES         .0 Platform Nintendo
2    MarioKart  Wii         1985    Racing Nintendo
3  WiiSports Resort  Wii         .0    Sports Nintendo
4  PokemonRed/Pokemon Blue GB         2008 RolePlaying Nintendo
      NA_Sales EU_Sales Sales Other_Sales Global_Sales Critic_Score
0    41.3    28.96    3.77    8.45    .0    82.53    76.0
1     6         3.58    6.81    0.77    1994    40.24    NaN
2    29.0    12.76    3.79    3.29    .0    35.52    82.0
3     8         10.93    3.28    2.95         32.77    80.0
4    15.6     8.89    10.22    1.00         31.37    NaN
      Critic_Count User_Score User_Count Developer Rating
0     1         8    322.0 Nintendo E    51.0
1    11.2        NaN        NaN    NaN NaN    NaN
2     7         8.3    709 Nintendo E    73.0
3         8         .0    do E    73.0
4        NaN    192.0 Nintendo NaN NaN    NaN
      .0    do

```

```
[4]: steam_df.columns
```

```

[4]: Index(['AppID', 'Name', 'Release date', 'Estimated owners', 'Peak CCU',
'Required age', 'Price', 'DiscountDLC count', 'About the game',
'Supported languages', 'Full audio languages', 'Reviews',
'Header image', 'Website', 'Support url', 'Support email', 'Windows',
'Mac', 'Linux', 'Metacritic score', 'Metacritic url', 'User score',
'Positive', 'Negative', 'Score rank', 'Achievements', 'Recommendation',
'Notes', 'Average playtime forever', 'Average playtime two weeks',
'Median playtime forever', 'Median playtime two weeks', 'Developers',
'Publishers', 'Categories', 'Genres', 'Tags', 'Screenshots', 'Movies'],

```

```
dtype='object')
```

```
[6]: sales_df.columns
```

```
[6]: Index(['Name', 'Platform', 'Year_of_Release', 'Genre', 'Publisher', 'NA_Sales',  
        'EU_Sales', 'JP_Sales', 'Other_Sales', 'Global_Sales', 'Critic_Score',  
        'Critic_Count', 'User_Score', 'User_Count', 'Developer', 'Rating'],  
        dtype='object')
```

```
[11]: steam_df.info()
```

```
<class 'pandas.core.frame.DataFrame'>  
Index: 122611 entries, 2539430 to 4141790  
Data columns (total 39 columns):  
#   Column                Non-Null Count  Dtype  
---  ---  
0   ---                  122611 non-null   object  
1   AppID                 null              t  
2   Date                  122611 non-null   object  
3   Name                  null              t  
4   ReleaseDate           122611 non-null   object  
5   PeakCCU               null              t  
6   Required              122611 non-null   int64  
7   Price                 null              int64  
8   Abscort               122611 non-null   float  
9   DLC                   null              64  
10  Support languages     122611 non-null   int64  
11  ed                    122611 non-null   int64  
12  Field of view         122611 non-null   object  
13  Website              496 non-null      t  
14  Support url           122611 non-null   object  
15  Support email         122611 non-null   t  
16  Windows               122611 non-null   object  
17  Mac                   122611 non-null   t  
18  Linux                 122611 non-null   object  
19  Metacritic            122611 non-null   t  
20  Metacritic score      122611 non-null   object  
21  Userscore             122611 non-null   t  
22  Positive              122611 non-null   t  
23  Negative              null              object  
24  Score                 122611 non-null   t  
25  Achievement            null              object  
26  s                     122611 non-null   t  
27  Recommendations       611 non-null      object  
28  tions Notes           122611 non-null   t  
29  Average playtime fore 122611 non-null   bool  
30  Average playtime two 122611 non-null   bool  
31  Median playtime fore  122611 non-null   bool  
32  Median playtime two  122611 non-null   int64  
33  5                     int64  
34  int64  
35  int64  
36  float
```

```

3   Developers      114174 non-objec
2   rs              null      t
3   Publishers      113702 non-objec
3   s              null      t
3   Categories      833458 non-objec
4   s Genres        1161593 non-null
3   Tags            114198 non-objec
5   Screenplay      1161593 non-null
3   Memory usage: 35.0+ MB      objec
6   Movies          t

```

```

[12]: sales_df.info()      objec

```

```

7
<class
3   pandas.core.frame.DataFrame      float
8   > RangeIndex: 16719      64
# Columns: 0 to 16719 Data Dtype
columns (total 16 columns):-----
0 Name      1167 non-null      objec
1 Platform   117 non-null      t
2 Year_of_Release 1167 non-null      objec
3 Genre      110 non-null      t
4 Publisher   1164 non-null      float
5 NA_Sales    50 non-null      64
6 EU_Sales    1167 non-null      objec
7 JP_Sales    117 non-null      t
8 Other_Sales 1166 non-null      objec
9 Global_Sales 65 non-null      t
10 Critic_Score 167 non-null float
11 Critic_Count 19 non-null 64
1 User_Sc      167 non-null float
2 ore          19 non-null 64
1 User_Co      167 non-null float
3 unt          19 non-null 64
1 Developer    1167 non-null float
Memory usage: 2.09 MB      64
1 Rating       167 float

```

```

[13]: steam_df.duplicated().sum() 64

```

```

[13]: 7      813      float
      7      64

```

```

[15]: steam_df = steam_df.drop_duplicates()

```

```

[16]: steam_df.duplicated().sum()      objec
      15      t

```

```

[16]: 0      759      float
      0      64

```

```

[14]: sales_df.duplicated().sum()      objec
      96      t
      995      objec
      0      t

```

[14]: 0

Now that I've made sure that both files imported successfully, I've just been inspecting them to see if there are duplicates and removing those. Then I'll move on to NAN and make sure everything is clean before analysis.

```
[10]: # Test Steam API Data Collection
appid = 1245620 # Elden Ring

url = f"https://store.steampowered.com/api/appdetails?appids={appid}"

response = requests.get(url)
data = response.json()
data
```

```
[10]: {'1245620': {'success': True,
  'data': {'type': 'game',
    'name': 'ELDEN RING',
    'steam_appid': 1245620,
    'required_age': '16',
    'is_free': False,
    'controller_support': 'full',
    'dlc': [2778590, 2778580],
    'detailed_description': '<h1>ELDEN RING Shadow of the Erdtree
Edition</h1><p><span
class="bb_img_ctn"><video
class="bb_img" autoplay muted
loop
playsinline
poster="https://shared.akamai.steamstatic.com/store_item_assets
/steam/apps/1245620/extras/6e68c7bbe71819c202f190579f8d4
f12.poster.avif?t=176788 3716" width=616 height=346 ><source
src="https://shared.akamai.steamstatic.com/s
tore_item_assets/steam/apps/1245620/extras/6e68c7bbe71819
c202f190579f8d4f12.webm ?t=1767883716" type="video/webm;
codecs=vp9"><source
src="https://shared.akamai.
steamstatic.com/store_item_assets/steam/apps/1245620/extras
/6e68c7bbe71819c202f1 90579f8d4f12.mp4?t=1767883716"
type="video/mp4"></video></span><br>ELDEN RING Shadow of
the Erdtree Edition includes:<br><ul class="bb_ul"><li>ELDEN
RING<br></li><li>ELDEN RING Shadow of the Erdtree
expansion</li></ul></p><br><h1>ELDEN RING Shadow of the
Erdtree Deluxe Edition</h1><p><span
class="bb_img_ctn">
<video class="bb_img" autoplay muted
loop
playsinline
poster="https://shared.akamai.steamstatic.com/store_item_assets
/steam/apps/1245620/extras/0856a96484c4fb0a35eee2db28320
da2.poster.avif?t=176788 3716" width=616 height=346 ><source
src="https://shared.akamai.steamstatic.com/s
tore_item_assets/steam/apps/1245620/extras/0856a96484c4fb0
a35eee2db28320da2.webm ?t=1767883716" type="video/webm;
codecs=vp9"><source
src="https://shared.akamai.
steamstatic.com/store_item_assets/steam/apps/1245620/extras
```


Game</h1>THE CRITICALLY ACCLAIMED FANTASY ACTION

RPG
<i>Rise, Tarnished, and be guided by grace to brandish the power of the Elden Ring.</i><h2 class="bb_tag" >•

A Breathtaking World Full of Excitement and Mystery</h2>The Lands Between are part of a vast continent where magnificent open fields and huge dungeons with complex and three-dimensional designs are seamlessly connected. As you explore, the joy of discovering unknown and overwhelming threats awaits you.
Mastery of the terrain and knowledge of its secrets can help you overcome enemies and defeat formidable bosses or lead invading players into traps.

<video class="bb_img" autoplay muted loop playsinline crossorigin="anonymous" poster="h

ttps://shared.akamai.steamstatic.com/store_item_assets/steam/apps/1245620/extras

/b2d503549e33e6603c86b6bd7babdb38.poster.avif?

t=1767883716" width=780 height=320 ><source

src="https://shared.akamai.steamstatic.com/store_item_assets/steam/apps

/1245620/extras/b2d503549e33e6603c86b6bd7babdb38.webm?

t=1767883716" type="video/webm; codecs=vp9"></video>

<h2 class="bb_tag" >• Defeat Challenging Foes in Intense Combat</h2>Combat in ELDEN RING is simple to learn yet offers hidden depths of mastery. As you seek to become the Elden Lord,

you'll need to explore the balance between attacking and avoiding damage, use a wide variety of weapons, spells, & summons, and perfect your timing to take advantage of your opponents' weaknesses.

<video class="bb_img" autoplay muted loop playsinline crossorigin="anonymous" poster="h

ttps://shared.akamai.steamstatic.com/store_item_assets/steam/apps/1245620/extras

/61c24948423742d3f67e094fe7be4119.poster.avif?

t=1767883716" width=780 height=320 ><source

src="https://shared.akamai.steamstatic.com/store_item_assets/steam/apps

/1245620/extras/61c24948423742d3f67e094fe7be4119.webm?

t=1767883716" type="video/webm; codecs=vp9"></video>

<h2 class="bb_tag" >• Create and Build Your Own Character</h2>In addition to customizing the appearance of your character, there are countless ways to combine the weapons, armor, usable items, and magic that you equip. You can develop your character according to your play style.
No matter if you prefer bold physical confrontation, tactical spellcasting, or the subtle art of stealth, you'll be able to find gear that supports your choices.

<span

class="bb_img_ctn"><video class="bb_img" autoplay muted loop playsinline crossorigin="anonymous" poster="https://shared.a

kamai.steamstatic.com/store_item_assets/steam/apps/1245620/extras/bb8df835b8ac3e

772150d98c157e0c41.poster.avif?

t=1767883716" width=780 height=320 ><source src="

c.com/store_item_assets/steam/apps/1245620/extras/8b612e65bf3ad3c2e81.webm?t=1767883716" type="video/webm; codecs=vp9"></video>
class="bb_tag" >• Play Alongside a Massive Worldwide Community</h2>The community is massive and active. Your friends may already be among them. play with up to two other Tarnished as your cooperative teammates, either inviting them using a shared password or by summoning them from a pool of community members.
There are also ample opportunities to face off against other players, either through co-op invasions, invited duels, or the many PvP battle options available in the three Colosseums.

<video class="bb_img" autoplay muted loop playsinline crossorigin="anonymous" poster="https://shared.akamai.steamstatic.com/store_item_assets/steam/apps/1245620/extras/8bed670f075526277b71e13a1c614f51.webm?t=1767883716" width=780 height=320 ><source src="https://shared.akamai.steamstatic.com/store_item_assets/steam/apps/1245620/extras/8bed670f075526277b71e13a1c614f51.webm?t=1767883716" type="video/webm; codecs=vp9"></video>
'about_the_game': 'THE CRITICALLY ACCLAIMED FANTASY ACTION RPG
<i>Rise, Tarnished, and be guided by grace to brandish the power of the Elden Ring.</i><h2 class="bb_tag" >• A Breathtaking World Full of Excitement and Mystery</h2>The Lands Between are part of a vast continent where magnificent open fields and huge dungeons with complex and three-dimensional designs are seamlessly connected. As you explore, the joy of discovering unknown and overwhelming threats awaits you.
Mastery of the terrain and knowledge of its secrets can help you overcome enemies and defeat formidable bosses or lead invading players into traps.

<video class="bb_img" autoplay muted loop playsinline crossorigin="anonymous" poster="https://shared.akamai.steamstatic.com/store_item_assets/steam/apps/1245620/extras/b2d503549e33e6603c86b6bd7babdb38.poster.avif?t=1767883716" width=780 height=320 ><source src="https://shared.akamai.steamstatic.com/store_item_assets/steam/apps/1245620/extras/b2d503549e33e6603c86b6bd7babdb38.webm?t=1767883716" type="video/webm; codecs=vp9"></video><h2 class="bb_tag" >• Defeat Challenging Foes in Intense Combat</h2>Combat in ELDEN RING is simple to learn yet offers hidden depths of mastery. As you seek to become the Elden Lord, you'll need to explore the balance between attacking and avoiding damage, use a wide variety of weapons, spells, & summons, and perfect your timing to take advantage of your opponents' weaknesses.

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```

```
[24]: sample_appid = steam_df["AppID"].iloc[0]
```

```
sample_appid
```

```
[24]: 'Black Dragon Mage Playtest'
```

```
[17]: steam_df.isnull().sum().sort_values(ascending=False)
```

```

[17]: Movies                122604
ScoreRank                  122564
Metacritic                  118348
Reviews                    110534
Notes                      100146
Website                    72934
Supporturl                  68463
Tags                       39258
Supportemail                22256
Categories                  8953
AbuseReports                8908
Publishers                  8449
Developers                  8432
Genres                     8413
Screenshots                 6018
Headerimage                 81
AppID                       1
Mac                         0
Linux                       0
Releasedate                 0
EstimatedOwners             0

```

```

PeakCCU                0
Requiredage            0
Price                  0
DiscountDLCcount       0
Medianplaytittwo weeks 0
me forever            0
Medianplaytittwo weeks 0
me forever            0
Supported languages    0
Recommendation         0
Averageplaytime        0
Fullaudiolanguages     0
es Negative           0
Positive               0
Userscore              0
Windows Name          0
Metacriticscore        0
dtype: int64          0

```

```
[18]: sales_df.isnull().sum().sort_values(ascending=False)
```

```

[18]: User_Count      9129
Critic_Score      8582
Critic_Count      8582
Rating            6769
User_Score        6704
Developer         6623
Year_of_Rel       269
ease              54
Publisher          2
Name Genre        2
Platform          0
NA_Sales          0
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JP_Sales          0
Other_Sales       0
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dtype: int64

```

```
[23]: steam_df.columns
```

```

[23]: Index(['AppID', 'Name', 'Release date', 'Estimated owners', 'Peak CCU',
            'Required age', 'Price', 'DiscountDLC count', 'About the game',
            'Supported languages', 'Full audio languages', 'Reviews',
            'Header image', 'Website', 'Support url', 'Support email', 'Windows',
            'Mac', 'Linux', 'Metacritic score', 'Metacritic url', 'User score',

```

```
'Positive', 'Negative', 'Score rank', 'Achievements', 'Recommendation',
'Notes', 'Average playtime forever', 'Average playtime two weeks',
'Median playtime forever', 'Median playtime two weeks', 'Developers',
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```

```
[26]: steam_df["Release date"] = pd.to_datetime(
    steam_df["Release date"],
    errors="coerce"
)

steam_df["Release year"] = (
    steam_df["Release date"].dt.year
)
```

C:\Users\ayden\AppData\Local\Temp\ipykernel_27360\3847174024.py:1: UserWarning: Could not infer format, so each element will be parsed individually, falling back to `dateutil`. To ensure parsing is consistent and as-expected, please specify a format.

```
steam_df["Release date"] = pd.to_datetime(
```

```
[27]: def owner_midpoint(owner_range):

    try:
        low, high = owner_range.split(" - ")

        return (int(low) + int(high)) / 2

    except:
        return np.nan
```

```
[28]: steam_df["Estimated owners numeric"] = (
    steam_df["Estimated owners"]
    .apply(owner_midpoint)
)
```

```
[29]: steam_df["Review ratio"] = (
    steam_df["Positive"] /
    (steam_df["Positive"] + steam_df["Negative"])
)
```

```
[30]: sample_appids = (
    steam_df["App
    pID"]
    .dropna()
    .head(50)
```

```
[52]: def get_game_storage(appid):

    url = f"https://store.steampowered.com/api/appdetails?appids={appid}"

    try:
        response = requests.get(url, timeout=10)
        data = response.json()
        app_data = data.get(str(appid), {})

        if not app_data.get("success"):
            return {
                "AppID": appid,
                "install_size_gb": None
            }

        game = app_data.get("data", {})

        if game is None:
            return {
                "AppID": appid,
                "install_size_gb": None
            }

        requirements = game.get("pc_requirements")

        storage_size = extract_storage_size(requirements)

        return {
            "AppID": appid,
            "install_size_gb": storage_size
        }

    except Exception as e:
        print("Error:", appid, e)
        return {
            "AppID": appid,
            "install_size_gb": None
        }
```

```
[47]: def extract_storage_size(requirements_text):
```

```
    if requirements_text is None:
        return None

    text = str(requirements_text)
```

```

# Specifically search for storage-related lines
storage_match = re.search(
    r'Storage.*?(\d+)\s?GB',
    text,
    re.IGNORECASE
)
if
    storage_match:
        return float(storage_match.group(1))

return None

```

```

[54]: test_appid = 1245620

url = f"https://store.steampowered.com/api/appdetails?appids={test_appid}"
response = requests.get(url)
data = response.json()
data[str(test_appid)]["success"]

```

[54]: True

```

[55]: game = data[str(test_appid)]["data"]

game.get("pc_requirements")

```

```

[55]: {'minimum': '<strong>Minimum:</strong><br><ul class="bb_ul"><li>Requi
64-bit processor and operating system<br></li><li><strong>OS:</strong> V
10<br></li><li><strong>Processor:</strong> INTEL CORE I5-8400 or AMD P
3300X<br></li><li><strong>Memory:</strong> 12 GB
RAM<br></li><li><strong>Graphics:</strong> NVIDIA GEFORCE GTX 1060 3
RADEON RX 580 4 GB<br></li><li><strong>DirectX:</strong> Version
12<br></li><li><strong>Storage:</strong> 60 GB available
space<br></li><li><strong>Sound Card:</strong> Windows Compatible Aud
Device<br></li><li><strong>Additional Notes:</strong> </li></ul>',
'recommended': '<strong>Recommended:</strong><br><ul class="bb_ul"><
a 64-bit processor and operating system<br></li><li><strong>OS:</strong>
10/11<br></li><li><strong>Processor:</strong> INTEL CORE I7-8700K or A
3600X<br></li><li><strong>Memory:</strong> 16 GB
RAM<br></li><li><strong>Graphics:</strong> NVIDIA GEFORCE GTX 1070 8
RADEON RX VEGA 56 8 GB<br></li><li><strong>DirectX:</strong> Version
12<br></li><li><strong>Storage:</strong> 60 GB available
space<br></li><li><strong>Sound Card:</strong> Windows Compatible Aud
Device<br></li><li><strong>Additional Notes:</strong> </li></ul>'}
```



```
[56]: extract_storage_size(game.get("pc_requirements"))
```

```
[56]: 60.0
```

This section of the project uses the Steam Store API to gather install size information for video games using each game's unique Steam AppID. The API returns metadata and system requirement information for each game, including storage requirements that are often embedded within text or HTML formatting. Python functions and regular expressions are used to extract the storage size values and convert them into usable numerical data. The cleaned install size data is then merged back into the main dataset containing reviews, ownership estimates, and sales-related metrics. This process allows the project to analyze relationships between game install size, commercial success, and player reception. I had a bit of troubleshooting trying to get the API return to specifically get the storage requirements. At first it pulled nothing, then a 12, which I know isn't the full download size for Elden Ring (no way at all), that must be the RAM recommendation/requirement as listed in the requirements above. It was just grabbing anything with GB attached to it at first, so I had to edit that. Now it's finally working and pulled 60 GB for install size.

```
[57]: get_game_storage(1245620)
```

```
[57]: {'AppID': 1245620, 'install_size_gb': 60.0}
```

```
[62]: def get_game_storage(appid):

    url = f"https://store.steampowered.com/api/appdetails?appids={appid}"

    try:
        response = requests.get(url, timeout=10)
        data = response.json()
        app_data = data.get(str(appid), {})

        if not app_data.get("success"):
            return{
                "AppID": appid,
                "install_size_gb": None
            }

        game = app_data.get("data", {})

        if game is None:
            return{
                "AppID": appid,
                "install_size_gb": None
            }

        requirements = game.get("pc_requirements")

        storage_size = extract_storage_size(requirements)
```

```

    return {
        "AppID": appid,
        "install_size_gb": storage_size
    }

except Exception as e:
    print("Error:", appid, e)
    return {
        "AppID": appid,
        "install_size_gb": None
    }

```

```

[63]: steam_sample = (
    steam_df
    .dropna(subset=["AppID"])
    .sort_values("Positive", ascending=False)
    .head(100)
)

sample_appids = steam_sample["AppID"]

```

```

[64]: storage_results = []

for appid in sample_appids:

    result = get_game_storage(appid)

    print(result)

    storage_results.append(result)

    time.sleep(1)

```

```

Error: Counter-Strike 2 'NoneType' object has no attribute
'get'
{'AppID': 'Counter-Strike 2', 'install_size_gb': None}
Error: Dota 2 'NoneType' object has no attribute 'get'
{'AppID': 'Dota 2', 'install_size_gb': None}
Error: Grand Theft Auto V Legacy 'NoneType' object has no
attribute 'get' {'AppID': 'Grand Theft Auto V Legacy',
'install_size_gb': None}
Error: PUBG: BATTLEGROUNDS 'NoneType' object has no
attribute 'get' {'AppID': 'PUBG: BATTLEGROUNDS',
'install_size_gb': None}
Error: Terraria 'NoneType' object has no attribute 'get'
{'AppID': 'Terraria', 'install_size_gb': None}
Error: Tom Clancy's Rainbow Six® Siege X 'NoneType' object
has no attribute 'get'
{'AppID': 'Tom Clancy's Rainbow Six® Siege X',
'install_size_gb': None} Error: Garry's Mod 'NoneType'
object has no attribute 'get'

```

```

{'AppID': 'Garry's Mod', 'install_size_gb': None}
Error: Black Myth: Wukong 'NoneType' object has no attribute 'get'
{'AppID': 'Black Myth: Wukong', 'install_size_gb': None}
Error: Rust 'NoneType' object has no attribute 'get'
{'AppID': 'Rust', 'install_size_gb': None}
Error: Team Fortress 2 'NoneType' object has no attribute 'get'
{'AppID': 'Team Fortress 2', 'install_size_gb': None}
Error: ELDEN RING 'NoneType' object has no attribute 'get'
{'AppID': 'ELDEN RING', 'install_size_gb': None}
Error: Left 4 Dead 2 'NoneType' object has no attribute 'get'
{'AppID': 'Left 4 Dead 2', 'install_size_gb': None}
Error: Wallpaper Engine 'NoneType' object has no attribute 'get'
{'AppID': 'Wallpaper Engine', 'install_size_gb': None}
Error: Stardew Valley 'NoneType' object has no attribute 'get'
{'AppID': 'Stardew Valley', 'install_size_gb': None}
Error: Euro Truck Simulator 2 'NoneType' object has no attribute 'get'
{'AppID': 'Euro Truck Simulator 2', 'install_size_gb': None}
Error: The Witcher 3: Wild Hunt 'NoneType' object has no attribute 'get'
{'AppID': 'The Witcher 3: Wild Hunt', 'install_size_gb': None}
Error: HELLDIVERS 2 'NoneType' object has no attribute 'get'
{'AppID': 'HELLDIVERS 2', 'install_size_gb': None}
Error: Baldur's Gate 3 'NoneType' object has no attribute 'get'
{'AppID': 'Baldur's Gate 3', 'install_size_gb': None}
Error: Phasmophobia 'NoneType' object has no attribute 'get'
{'AppID': 'Phasmophobia', 'install_size_gb': None}
Error: Cyberpunk 2077 'NoneType' object has no attribute 'get'
{'AppID': 'Cyberpunk 2077', 'install_size_gb': None}
Error: Red Dead Redemption 2 'NoneType' object has no attribute 'get'
{'AppID': 'Red Dead Redemption 2', 'install_size_gb': None}
Error: Apex Legends 'NoneType' object has no attribute 'get'
{'AppID': 'Apex Legends', 'install_size_gb': None}
Error: Among Us 'NoneType' object has no attribute 'get'
{'AppID': 'Among Us', 'install_size_gb': None}
Error: Dead by Daylight 'NoneType' object has no attribute 'get'
{'AppID': 'Dead by Daylight', 'install_size_gb': None}
Error: ARK: Survival Evolved 'NoneType' object has no attribute 'get'
{'AppID': 'ARK: Survival Evolved', 'install_size_gb': None}
Error: The Forest 'NoneType' object has no attribute 'get'
{'AppID': 'The Forest', 'install_size_gb': None}
Error: PAYDAY 2 'NoneType' object has no attribute 'get'
{'AppID': 'PAYDAY 2', 'install_size_gb': None}
Error: Warframe 'NoneType' object has no attribute 'get'
{'AppID': 'Warframe', 'install_size_gb': None}
Error: Rocket League® 'NoneType' object has no attribute 'get'
{'AppID': 'Rocket League®', 'install_size_gb': None}
Error: Unturned 'NoneType' object has no attribute 'get'
{'AppID': 'Unturned', 'install_size_gb': None}
Error: Destiny 2 'NoneType' object has no attribute 'get'

```

```
{'AppID': 'Destiny 2', 'install_size_gb': None}
Error: War Thunder 'NoneType' object has no attribute 'get'
{'AppID': 'War Thunder', 'install_size_gb': None}
Error: Don't Starve Together 'NoneType' object has no attribute 'get'
{'AppID': "Don't Starve Together", 'install_size_gb': None}
Error: Lethal Company 'NoneType' object has no attribute 'get'
{'AppID': 'Lethal Company', 'install_size_gb': None}
Error: Valheim 'NoneType' object has no attribute 'get'
{'AppID': 'Valheim', 'install_size_gb': None}
Error: Dying Light 'NoneType' object has no attribute 'get'
{'AppID': 'Dying Light', 'install_size_gb': None}
Error: Monster Hunter: World 'NoneType' object has no attribute 'get'
{'AppID': 'Monster Hunter: World', 'install_size_gb': None}
Error: Portal 2 'NoneType' object has no attribute 'get'
{'AppID': 'Portal 2', 'install_size_gb': None}
```

KeyboardInterrupt

Traceback (most recent call last)

Cell In[64], line 11

7 print(result)

9 storage_results.append(result)

---> 11 time.sleep(1)

KeyboardInterrupt:

```
[60]: storage_df = pd.DataFrame(storage_results)

storage_df.head()
```

```
[60]:
```

	AppID	install_size_gb
0	Counter-Strike	None
1	Dota	None
2	GrandTheftAutoVLegacy	None
3	PUBG:ATTLEGROUNDS	None

```
[61]: storage_df["install_size_gb"].notna().sum()
```

[61]: 0

```
[65]: steam_df[["AppID", "Name", "Positive"]].head()
```

```
[65]:
```

	AppID	Name	Positive
25394	BlackDragonMage	Playtest Aug1,2023	0
30	SupiparaChapter	SpringHasCome! Jul29,2016	252
49635	MysterySolitaire	TheBlackRaven May6,2019	21
0	██████████	VtubeParanoiaOct31,2024	0
10344		MazeQuestVR Apr24,2025	0
00			
32921			
90			
36310			
80			

```
[66]: steam_sample = (
    steam_df
    .dropna(subset=["AppID"])
    .sort_values("Positive", ascending=False)
    .head(50)
)

sample_appids = steam_sample["AppID"]
```

```
[67]: sample_appids.head()
```

```
[67]: 730          Counter-Strike2
      570 271590          Dota2
      578080 GrandThe AutoLegacy
      105600 ft BattleLEGROUNDS
      AppID, dtype=object
      Terraria
      G:
```

```
[68]: steam_df.columns
```

```
[68]: Index(['AppID', 'Name', 'Release date', 'Estimated owners', 'Peak CCU',
            'Required age', 'Price', 'DiscountDLC count', 'About the game',
            'Supported languages', 'Full audio languages', 'Reviews',
            'Header image', 'Website', 'Support url', 'Support email', 'Windows',
            'Mac', 'Linux', 'Metacritic score', 'Metacritic url', 'User score',
            'Positive', 'Negative', 'Score rank', 'Achievements', 'Recommendation',
            'Notes', 'Average playtime forever', 'Average playtime two weeks',
            'Median playtime forever', 'Median playtime two weeks', 'Developers',
            'Publishers', 'Categories', 'Genres', 'Tags', 'Screenshots', 'Movies',
            'Release year', 'Estimated owners numeric', 'Review ratio'],
            dtype='object')
```

```
[69]: steam_df = steam_df.reset_index()
```

```
[70]: steam_df[["AppID", "Name"]].head()
```

```
[70]:
      AppID      Name
0      BlackDragon MagePlaytes Aug1,2023
1 Supipara - Chapter 1 Spring HasJcom20, 2016
2      MysterySolitaireTheBlackRaven May6,2019
3      0000000000000000 Paranoia Oct31,2024
4      MazeQuest VR Apr24,2025
```

```
[71]: steam_sample = (
    steam_df
    .dropna(subset=["AppID"])
    .sort_values("Positive", ascending=False)
    .head(50)
```

```
)

sample_appids = steam_sample["AppID"]

sample_appids.head()
```

```
[71]: 45509      Counter-Strike2
      4193      Dota2
      59122 GrandThe AutoLegacy
      8878      ft BATTLEGROUNDS
      10927      PUB      Terraria
      Name: AppID, dtype: object
```

```
[72]: steam_df.head()
```

```
[72]:   index      AppID      NameReleasedate
0  2539430  BlackDragonMapeytest  Aug1,2023      NaT
1  496350  SupiparaChapter1  SpringHasComeJul29,2016      NaT
2  1034400  MysterySolitaireTBackRaven  May6,2019      NaT
3  3292190  ??????VtuberParanoiaOct31,2024      NaT
4  3631080  MazeQuestVApr24,2025      NaT

      Estimated      Peak CCRequired agePriceDiscountDLC count
0  owners 0 0 0 0      0.0      0      0
1  1 0      0      0      65      0
2      0      5.2      0      0
3      0      4      0      1
4      0      4.9      0      0
      9
      As out the game
0      9      NaN
1  Springtime, April: when the cherry
trees come ... 2 Immerse yourself in the
most beloved, mystical... 3 synopsis
'Hello, I'm Hiyoro, a new YouTuber!' ... 4
Its not just a Maze; its a Quest! Enter
the ca...      Supported languages Full audio languages Reviews
0      []      []      N
1      ['English']      []      a
2  ['English','French','German','Russian']      []      N
3      ['Korean']      ['Korean']      N
4      ['English']      ['English']      a
      Header image      N
0      a
https://shared.akamai.steamstatic.com/
store_it...      1      N
https://shared.akamai.steamstatic.com/
store_it...      2      N
https://shared.akamai.steamstatic.com/
store_it...      29      N
a
N
```

3	https://shared.akamai.steamstatic.co								
4	m/store_it...								
	https://shared.akamai.steamstatic.co								
	m/store_it...	Website \							
0		NaN							
1	http://mangagamer.org/supipara								
2	https://www.facebook.com/8FloorGames/								
3		NaN							
4	https://www.realityexpanded.com/books-games								

	Supporturl	Supportemail	Windows
0	NaN	NaN	True
1	http://mangagamer.com	support@mangagamer.com	True
2	https://www.facebook.com/8FloorGames	support@8floor.net	True
3	NaN	yujingamesc@gmail.com	True
4	https://www.realityexpanded.com	support@realityexpanded.com	True

	Mac	Linux	Metacritic score	Metacritic	User score	Positive\
0	False	False	NaN	0	0	0
1	False	False	NaN	0	0	252
2	True	False	NaN	0	0	21
3	False	False	NaN	0	0	0
4	False	False	NaN	0	0	0

	Negative Score	Recommendations	Reviews	
0	0	NaN	0	0
1	3	NaN	0	231
2	3	NaN	0	0
3	0	NaN	19	0
4	0	NaN	0	0

	Notes \
0	N
1	a
2	N
3	The game includes the following elements. 1. G...
4	NaN
	N

	Average playtime for	Average playtime in two weeks
0	0	a 0
1	8	N 0
2	0	0
3	0	0
4	0	0

	Median playtime for	Median playtime in two weeks	Developer's
0	0	0	NaN

1	8	0	minori
2	0	0	SomerGames
3	0	0	□□□□□
4	0	0	RealityExpandedLLC

	Publishers	Categories \
0	NaN	NaN
1	MangaGan	Single-player, Steam Trading Cards, Steam Cloud, ...
2	8floor	Single-player, Family Sharing
3	□□□□□	Single-player, Steam Achievements, Family Sharing
4	Reality Expanded LLC	Single-player, VR Only, Steam Leaderboards, Famil...

	Genres	Tags \
0	NaN	NaN
1	Adventure	Adventure, Visual Novel, Anime, Cute
2	Casual	Casual, Card Game, Solitaire, Puzzle, Hidden Objec...
3	Casual, Indie, Simulation	NaN
4	Action, Early Access	NaN

	Screenshots	Movies	Release year \
0	https://shared.akamai.steamstatic.co	Na	N
1	m/store_it...	N	a
2	https://shared.akamai.steamstatic.co	Na	N
3	m/store_it...	N	N
4	https://shared.akamai.steamstatic.co	Na	a
	m/store_it...	N	N
	https://shared.akamai.steamstatic.co	Na	N
0	m/store_it...	NaN	NaN
1	https://shared.akamai.steamstatic.co	NaN	NaN
2	m/store_it...	NaN	0.875000
3		NaN	NaN
4	Estimated owners	NaN	Numeric NaN

```
[76]: steam_df = steam_df.rename(
      columns={"Real_AppID": "index"}
    )
```

```
[78]: sample_appids = steam_sample["index"].astype(int)
```

```
[79]: steam_sample = (
      steam_df
      .dropna(subset=["index"])
      .sort_values("Positive", ascending=False)
      .head(50)
    )

sample_appids = (
```



```
steam_sample["index"]  
    .astype(int)  
)
```

```
[80]: sample_appids.head()
```

```
[80]: 45509    730  
      4193     570  
      59122  271590  
      8878   578080  
      10927  105600  
      Name: index, dtype: int32
```

```
[81]: storage_results = []  
  
for appid in sample_appids:  
    result = get_game_storage(appid)  
    print(result)  
    storage_results.append(result)  
    time.sleep(1)
```

```
{'AppID': 730, 'install_size_gb':  
85.0} {'AppID': 570,  
'install_size_gb': 60.0} {'AppID':  
271590, 'install_size_gb': 125.0}  
{'AppID': 578080,  
'install_size_gb': 40.0} {'AppID':  
105600, 'install_size_gb': None}  
{'AppID': 359550,  
'install_size_gb': 65.0} {'AppID':  
4000, 'install_size_gb': 10.0}  
{'AppID': 2358720,  
'install_size_gb': 130.0} {'AppID':  
252490, 'install_size_gb': 45.0}  
{'AppID': 440, 'install_size_gb':  
15.0} {'AppID': 1245620,  
'install_size_gb': 60.0} {'AppID':  
550, 'install_size_gb': 13.0}  
{'AppID': 431960,  
'install_size_gb': 2.0} {'AppID':  
413150, 'install_size_gb': None}  
{'AppID': 227300,  
'install_size_gb': None} {'AppID':  
292030, 'install_size_gb': 50.0}  
{'AppID': 553850,  
'install_size_gb': 135.0} {'AppID':  
1086940, 'install_size_gb': 150.0}  
{'AppID': 739630,  
'install_size_gb': 21.0} {'AppID':  
1091500, 'install_size_gb': 70.0}
```

```

{'AppID': 381210,
 'install_size_gb': 50.0} {'AppID': 346110,
 'install_size_gb': 60.0}
{'AppID': 242760,
 'install_size_gb': 5.0} {'AppID': 218620,
 'install_size_gb': 83.0}
{'AppID': 230410,
 'install_size_gb': 75.0} {'AppID': 252950,
 'install_size_gb': 20.0}
{'AppID': 304930,
 'install_size_gb': 4.0} {'AppID': 1085660,
 'install_size_gb': 105.0}
{'AppID': 236390,
 'install_size_gb': 70.0} {'AppID': 322330,
 'install_size_gb': 3.0}
{'AppID': 1966720,
 'install_size_gb': 4.0} {'AppID': 892970,
 'install_size_gb': 1.0}
{'AppID': 239140,
 'install_size_gb': None} {'AppID': 582010,
 'install_size_gb': 52.0}
{'AppID': 620,
 'install_size_gb': 8.0} {'AppID': 1938090,
 'install_size_gb': None} {'AppID': 367520,
 'install_size_gb': 9.0}
{'AppID': 322170,
 'install_size_gb': None} {'AppID': 374320,
 'install_size_gb': 25.0}
{'AppID': 1097150,
 'install_size_gb': 2.0} {'AppID': 1623730,
 'install_size_gb': 40.0}
{'AppID': 108600,
 'install_size_gb': None} {'AppID': 960090,
 'install_size_gb': None}
{'AppID': 250900,
 'install_size_gb': 8.0} {'AppID': 548436,
 'install_size_gb': 85.0}
KeyboardInterrupt: Traceback (most recent call last)
In [81], line 11
      7 print(result)
      8 storage_results.append(result)
--> 11 time.sleep(1)
KeyboardInterrupt:
[82]: storage_df = pd.DataFrame(storage_results)
storage_df.head()

```

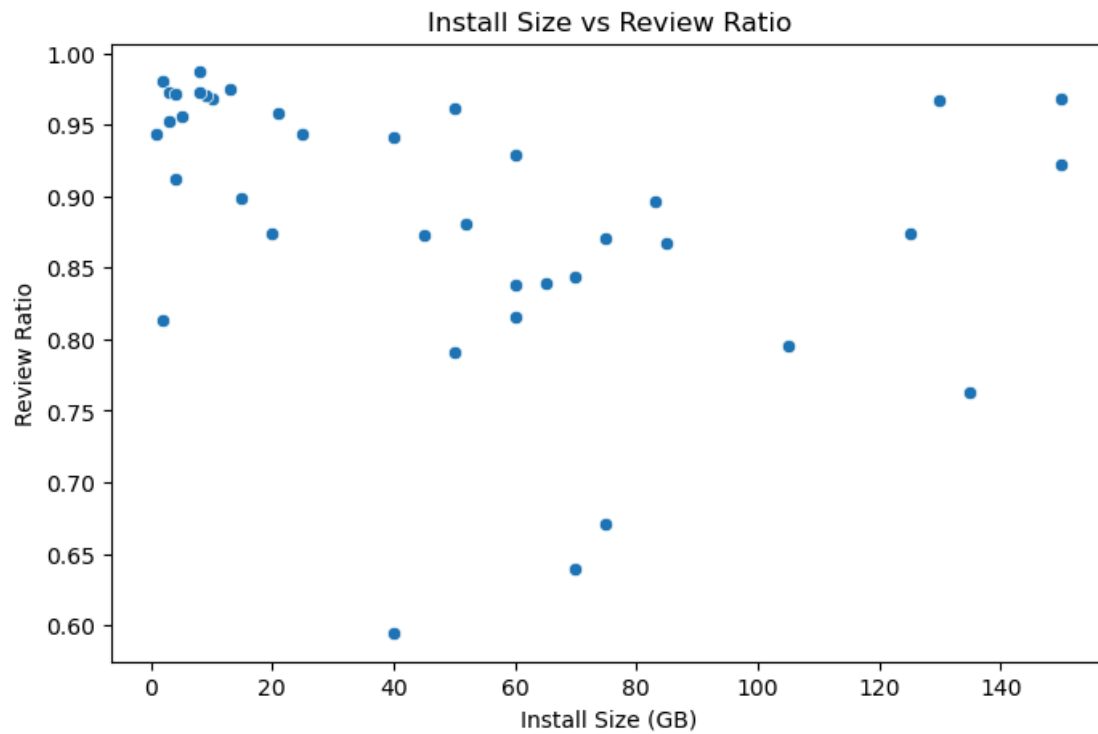
```

[82]: AppID install_size_gb
0    730          85.0
1    570          60.0
2  271590        125.0
3  578080          40.0
4  105600          NaN

```



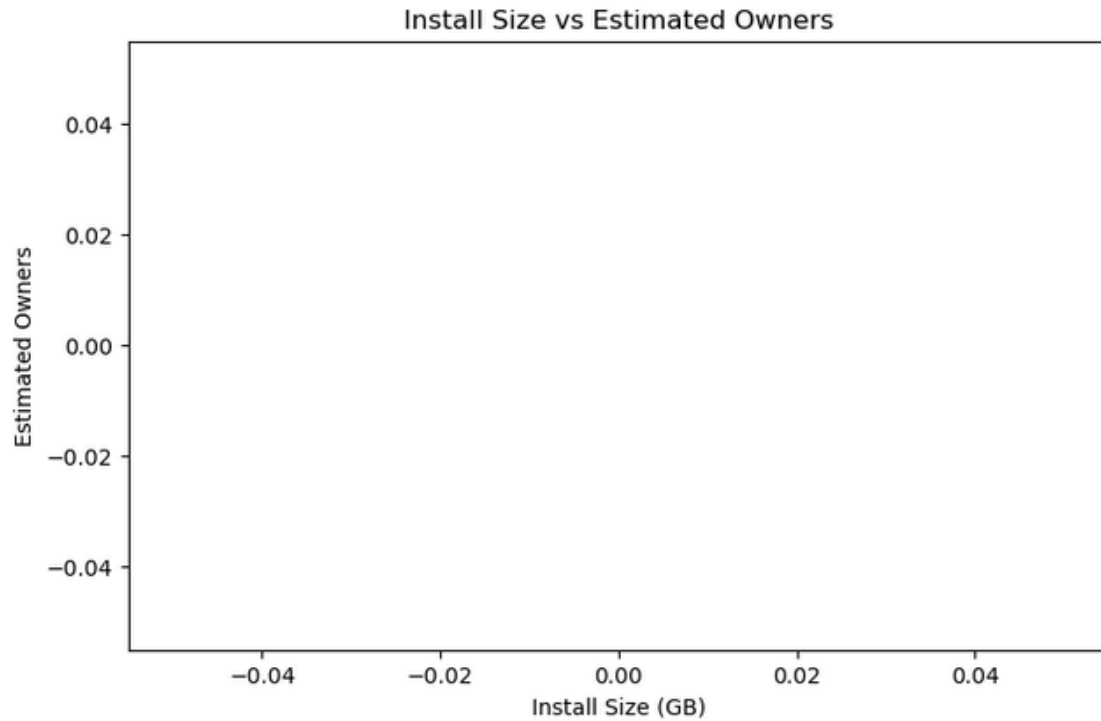
```
plt.show()
```



```
[87]: plt.figure(figsize=(8,5))

sns.scatterplot(
    data=steam_storage_df,
    x="install_size_gb",
    y="Estimated owners numeric"
)

plt.title("Install Size vs Estimated Owners")
plt.xlabel("Install Size (GB)")
plt.ylabel("Estimated Owners")
plt.show()
```



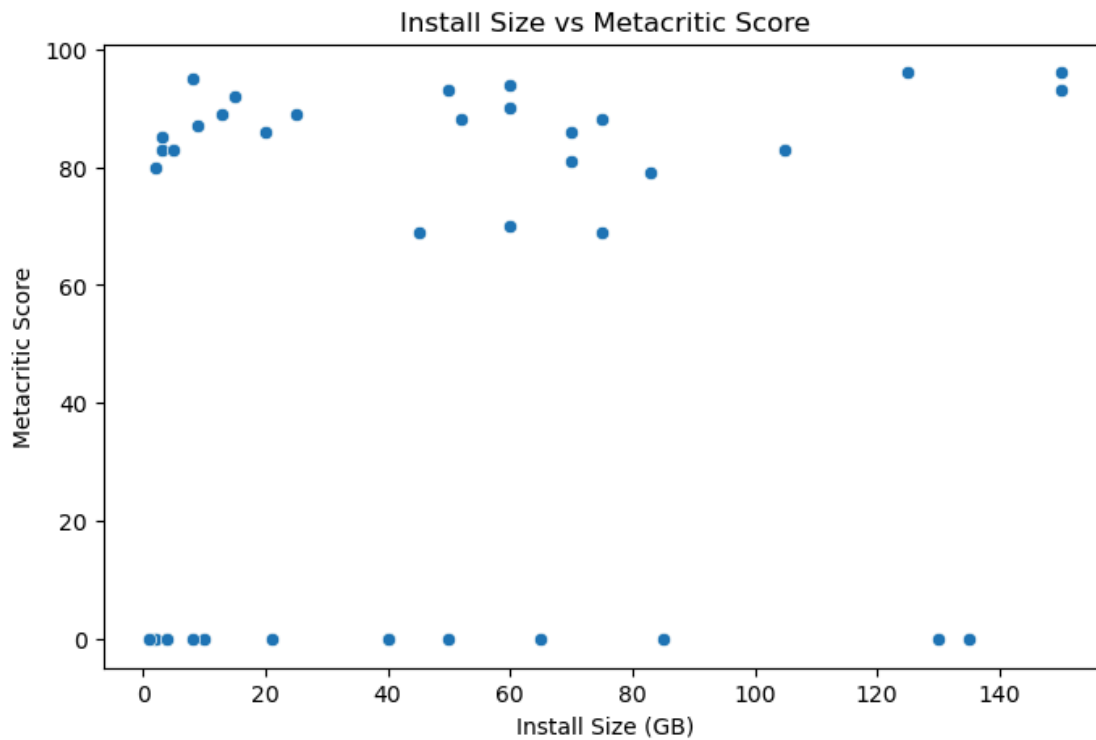
```
[92]: steam_df["Estimated owners"].head(10)
```

```
[92]: 0    0
      1    0
      2    0
      3    1
      4    0
      5    0
      6    8
      7    0
      8    0
      9    1
      Name: Estimated owners, dtype: int64
```

```
[88]: plt.figure(figsize=(8,5))

sns.scatterplot(
    data=steam_stora
    ge_df,
    x="install_size_g
)  b", y="Metacritic
    score"
plt.title("Install Size vs Metacritic Score")
```

```
plt.xlabel("Install Size (GB)")
plt.ylabel("Metacritic Score")
plt.show()
```



```
[89]: corr_df = steam_storage_df[[
    "install_size_gb",
    "Estimated owners numeric",
    "Review ratio",
    "Metacritic score"
]].corr()

corr_df
```

```
[89]:
```

	install_size_gb	Estimated owners numeric	Review ratio	Metacritic score
install_size_gb	1.000000	NaN	NaN	NaN
Estimated owners numeric	NaN	1.000000	NaN	NaN
Review ratio	-0.302169	NaN	1.000000	NaN
Metacritic score	0.135688	NaN	NaN	1.000000

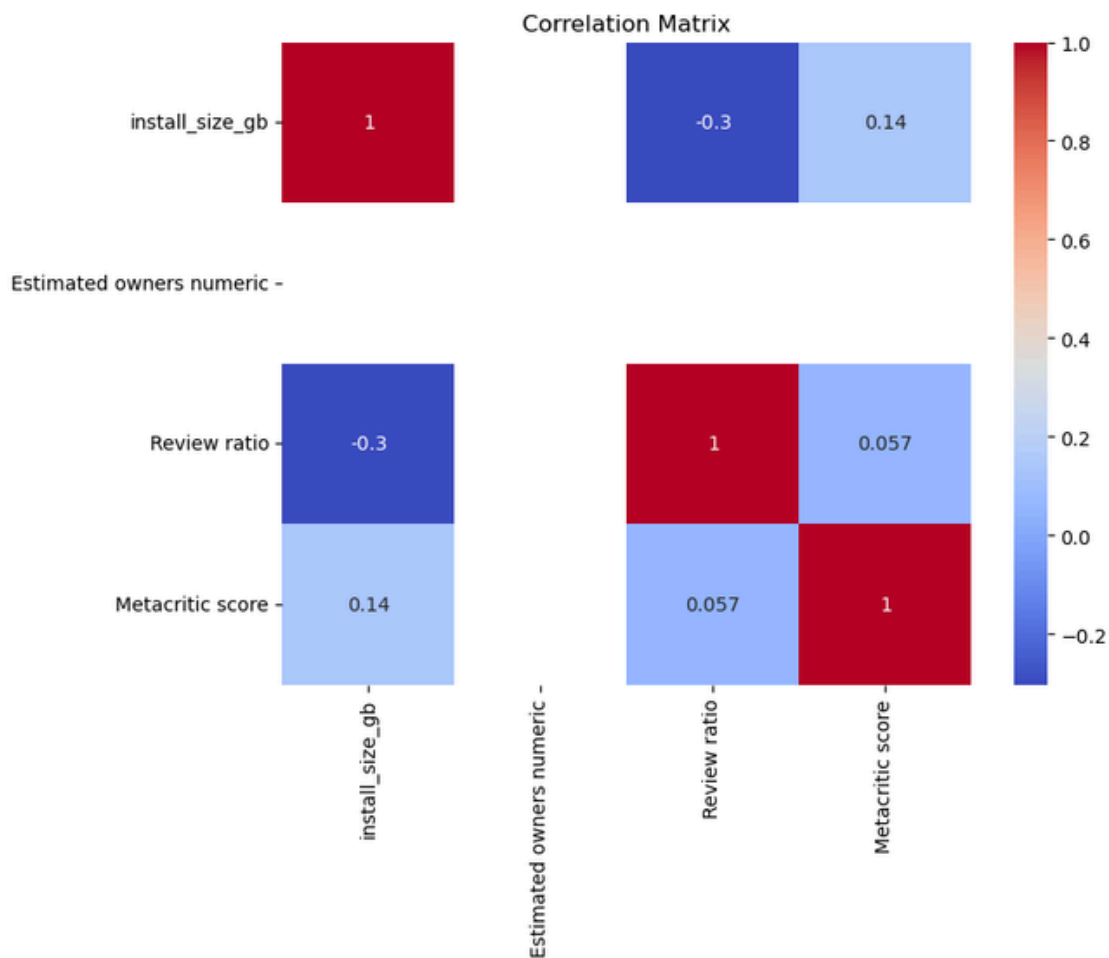
install_size_gb	-0.302169	0.135688
Estimated owners numeric	NaN	NaN
Reviewratio	1.000000	0.056757
Metacriticscore	0.056757	1.000000

```
[90]: plt.figure(figsize=(8,6))

sns.heatmap(
    corr_df,
    annot=True,
    cmap="coolwarm"
)

plt.title("Correlation Matrix")

plt.show()
```



```
[91]: largest_games = (
    steam_storage_df[[
        "AppID_x",
        "install_size_gb",
        "Estimated owners numeric"
    ]]
    .sort_values("install_size_gb", ascending=False)
    .head(10)
)

largest_games
```

```
[91]:
```

	AppID	install_size_gb	Estimated owners numeric
593	Baldur's Gate	150.	NaN
54	Red Dead Redemption 2	0	NaN
965	HELLDIVERS	135.0	NaN
70	Black Myth Wukong	130.0	N
652	Grand Theft Auto V Legacy	125.0	a
50	Destiny 2	105.0	N
441	Counter-Strike	85.0	N
34	PAYDAY2	83.0	a
591	Warframe	75.0	N
22	Apex Legends	75.0	NaN
287			a

```
[93]: ccu_df = steam_storage_df.dropna(
    subset=["install_size_gb", "Peak CCU"]
)

plt.figure(figsize=(8,5))

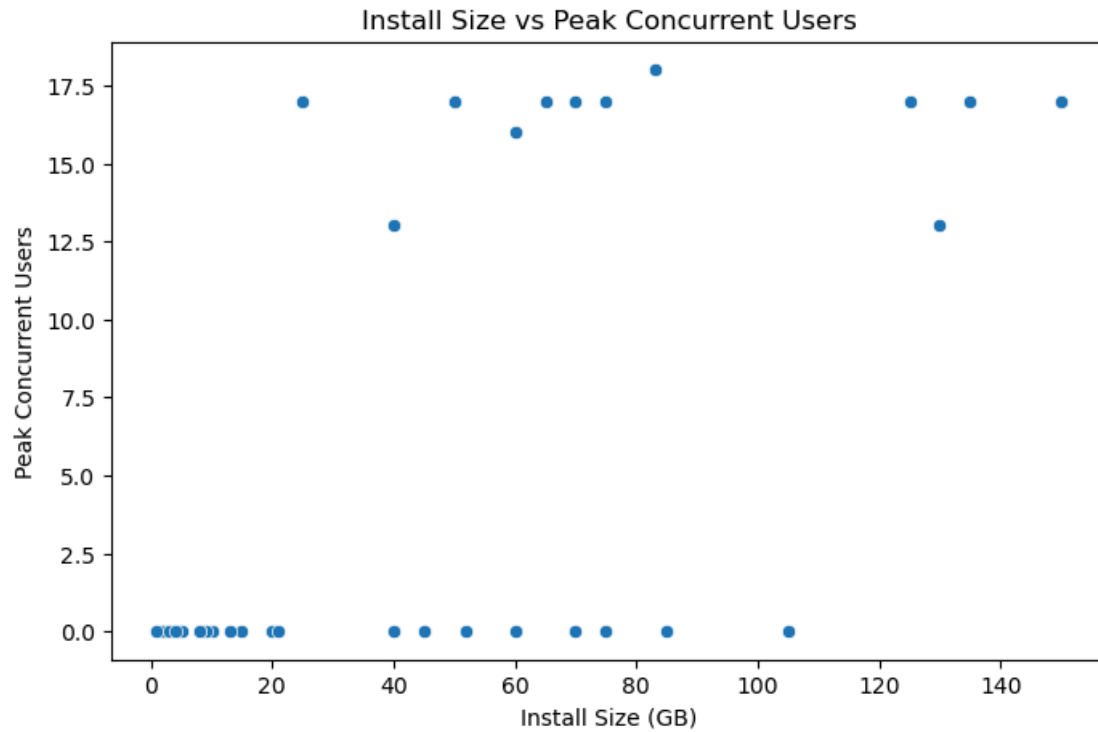
plt.scatterplot(
    x=ccu_df,
    y="install_size_gb",
    y="Peak CCU"
)

plt.title("Install Size vs Peak Concurrent Users")

plt.xlabel("Install Size (GB)")

plt.ylabel("Peak Concurrent Users")

plt.show()
```

```
[94]: rec_df = steam_storage_df.dropna(
        subset=["install_size_gb", "Recommendations"]
    )

    plt.figure(figsize=(8,5))

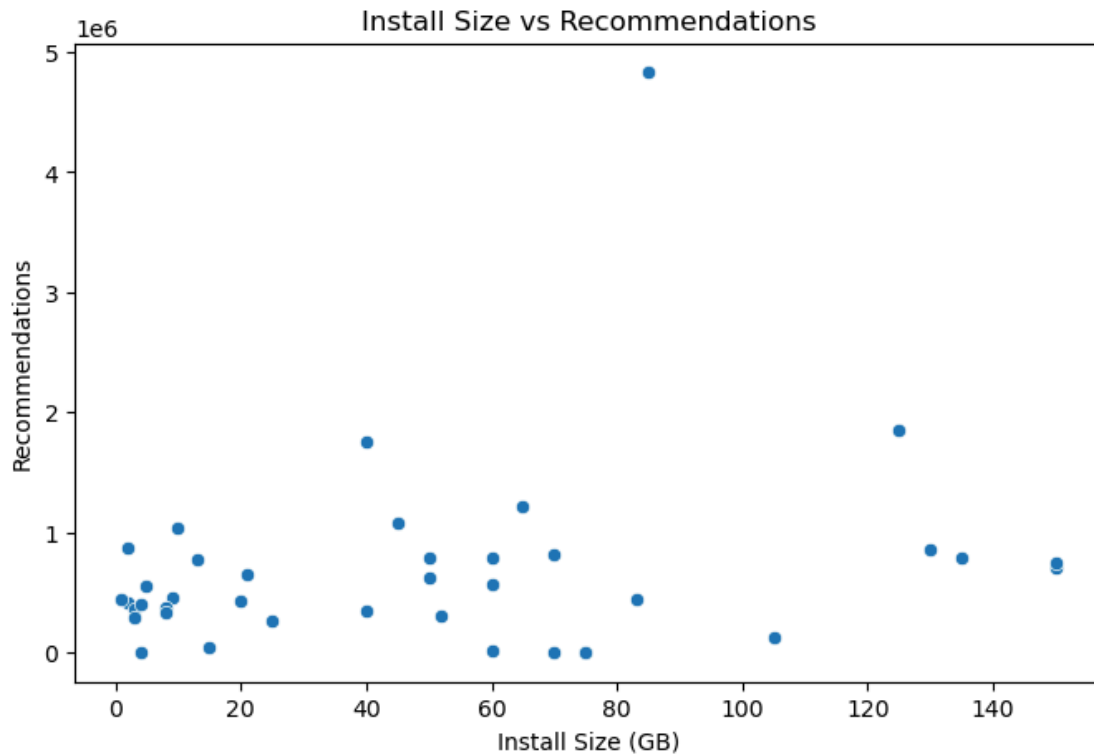
    sns.scatterplot(
        data=rec_df,
        x="install_size_gb",
        y="Recommendations"
    )

    plt.title("Install Size vs Recommendations")

    plt.xlabel("Install Size (GB)")

    plt.ylabel("Recommendations")

    plt.show()
```



```
[100]: meta_df = steam_storage_df[
    (steam_storage_df["install_size_gb"].notna()) &
    (steam_storage_df["Metacritic score"] > 0)
]
```

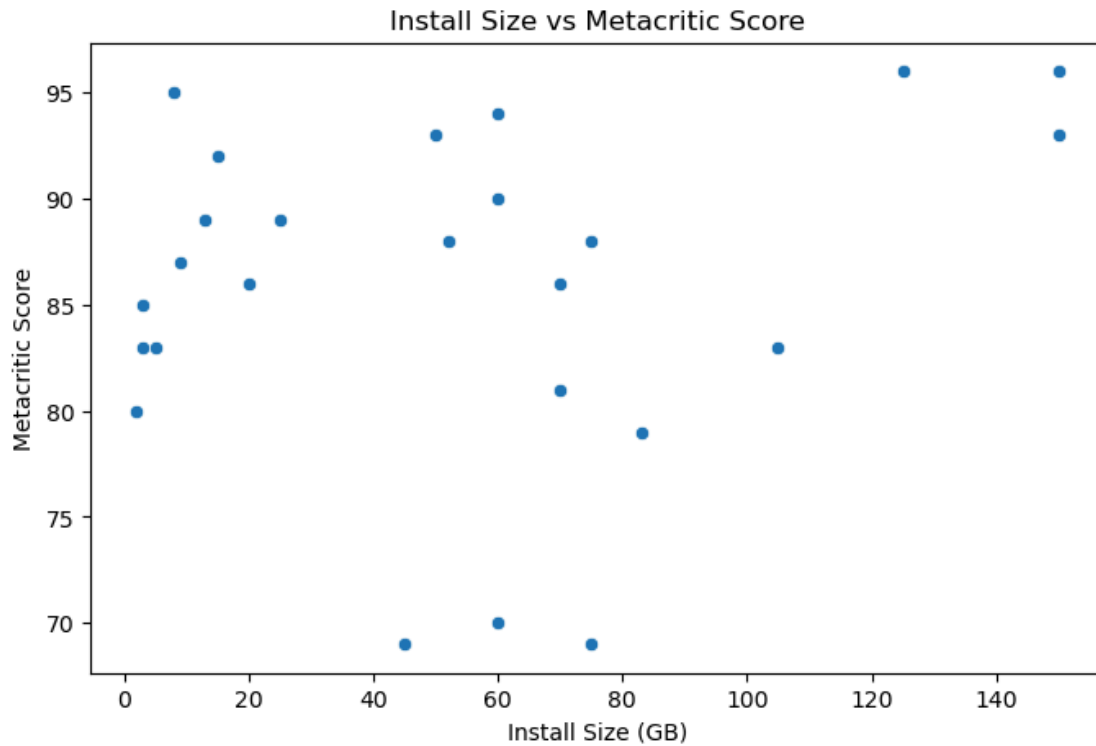
```
[104]: plt.figure(figsize=(8,5))

sns.scatterplot(
    data=meta_df,
    x="install_size_
    gb",
    y="Metacritic
    score"
)
plt.title("Install Size vs Metacritic Score")

plt.xlabel("Install Size (GB)")

plt.ylabel("Metacritic Score")

plt.show()
```



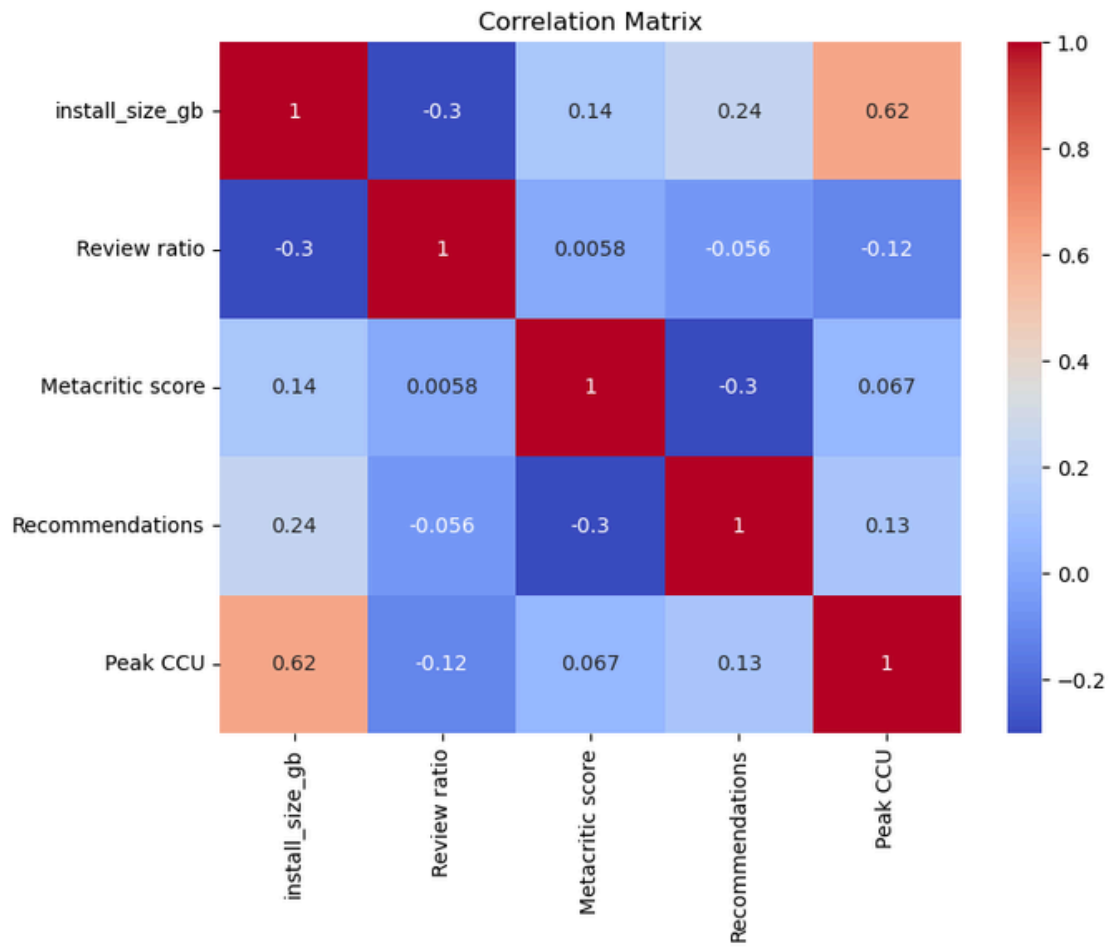
```
[102]: corr_df = steam_storage_df[[
    "install_size_gb",
    "Review ratio",
    "Metacritic score",
    "Recommendations",
    "Peak CCU"
]].dropna()
```

```
[103]: plt.figure(figsize=(8,6))

sns.heatmap(
    corr_df.corr(),
    annot=True,
    cmap="coolwarm"
)

plt.title("Correlation Matrix")

plt.show()
```



[:

Milestone 3 Audience Questions and Answers

Why was Steam chosen as the primary platform?

Steam provides one of the largest publicly available collections of PC gaming data, including reviews, recommendations, engagement metrics, and game metadata.

How accurate are the install size values?

The values were extracted from Steam system requirements and represent reasonable estimates, although updates and DLC can change storage requirements over time.

Why were some games missing install size data?

Some Steam listings contained incomplete or inconsistent system requirements, which resulted in missing values during extraction.

Does install size vary by genre?

Yes. Larger install sizes are commonly associated with open-world, simulation, and AAA action titles.

Could updates change install size values?

Yes. Patches and downloadable content frequently increase storage requirements after release.

Why use engagement metrics instead of sales figures?

Reliable sales figures are difficult to obtain, while recommendations, reviews, and peak concurrent users are publicly available and provide useful indicators of popularity.

Would console data change the results?

Possibly. Console ecosystems have different player behaviors and hardware limitations that may influence the findings.

How could review bombing affect the analysis?

Review bombing can temporarily distort review scores, although multiple engagement metrics were used to reduce its impact.

Could machine learning improve future versions of the project?

Yes. Predictive models could be used to estimate engagement or review performance based on install size and other game characteristics.

What additional variables could strengthen the analysis?

Genre, development budget, franchise status, DLC content, release year, and sales figures could provide additional insight.